DEVELOPMENT OF A PAY-AS-YOU-GO BATTERY-CHARGING SYSTEM FOR ELECTRIC MOBILITY IN NAIROBI, KENYA

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ABSTRACT

Abstract

The adoption of electric mobility in Nairobi is limited by high upfront costs, insufficient charging infrastructure, and uncertainties related to battery usage and financing (Oginga Martins, 2025). Pay-as-you-go (PAYG) models offer a promising approach to reduce financial barriers by linking energy access to incremental payments. However, the technical implementation of such systems requires reliable integration of payment platforms, software infrastructure, and physical hardware (Barry & Creti, 2020).

This work presents the development of a PAYG battery charging system for electric two-wheelers in collaboration with eWAKA. The system integrates a frontend interface, a Python-based backend, and a mobile payment function, enabling charging processes to be activated and controlled. The payment functionality was successfully implemented using the Daraja API, allowing to trigger charging sessions and the backend automatically terminate them once the purchased energy was consumed. The system was developed iteratively and tested in a controlled environment.

Two different approaches to enabling charging were investigated. A hardware-based solution for e-bikes was implemented as a prototype, demonstrating that charging control can be achieved independently of existing battery systems. In parallel, an API-based solution for e-motorbikes was developed, utilizing communication capabilities of smart batteries to enable direct control of charging without additional hardware.

The results show that both approaches are technically feasible, but differ significantly in terms of scalability and system complexity. While the hardware-based solution offers flexibility, it requires further development for large-scale deployment. The API-based approach provides a more efficient and cost-effective solution but introduces dependencies on battery manufacturers and external systems.

Overall, this work demonstrates the potential of PAYG charging systems to support the adoption of electric mobility in Nairobi and highlights the importance of system integration, scalability, and reliable communication for successful implementation.

1 INTRODUCTION

1.1 Electromobility in Africa

Electromobility is gaining increasing attention in Nairobi, Kenya, as a potential solution to address rising fuel costs, urban air pollution, and greenhouse gas emissions (Oginga Martins, 2025). In many African cities, including Nairobi, transport systems are dominated by two- and three-wheelers, particularly motorbikes, which are widely used for both passenger and delivery services (e.g., boda-boda transport) (Sietchiping *et al.*, 2012). These vehicles represent a significant share of daily urban mobility and are therefore a key entry point for electrification.

In Nairobi, motorbikes are commonly used for last-mile transport, ride-hailing services, and logistics applications such as food delivery and parcel distribution. From an economic perspective, electric two-wheelers often offer lower total cost of ownership compared to internal combustion engine (ICE) vehicles. While the upfront investment for EVs is generally higher, operational costs are significantly reduced due to lower electricity prices compared to petrol and reduced maintenance requirements, as electric drivetrains have fewer moving parts (Dankers, 2024). Studies have also found that switching to e-mobility can reduce fuel and maintenance costs by up to 40% (Pawlak *et al.*, 2023). The lower operating and maintenance costs are a common reason why drivers choose an EV over an ICE vehicle (Dankers, 2024).

However, the adoption of electric mobility in Nairobi faces several barriers. These include limited charging infrastructure, high initial investment costs, and uncertainties related to battery lifetime and financing models (Oginga Martins, 2025). To address these barriers, alternative business models such as battery swapping and PAYG systems have been proposed. These models aim to reduce upfront costs and shift expenses to a usage-based structure, making electric mobility more accessible for low- and middle-income users (Mwangi *et al.*, 2025).

Overall, the transition to electric mobility in Nairobi is strongly linked to the specific characteristics of the local transport system, where motorbikes and e-bikes play a central role in both economic activity and daily life. Therefore, scalable and cost-effective solutions tailored to these vehicle types are essential for enabling widespread adoption.

1.1.1 Battery swapping

Battery swapping has emerged as one of the most prominent business models for enabling electric mobility in Sub-Saharan Africa (SSA), particularly for two-wheeler applications such as motorbikes. Instead of charging a battery within the vehicle, users exchange a depleted battery for a fully charged one at dedicated swapping stations. This approach significantly reduces vehicle downtime and eliminates the need for long charging periods, which is a critical factor for commercial drivers who rely on continuous operation for their income (Simwaba & Qutieshat, 2026).

In the context of Nairobi, battery swapping system (BSS) is especially relevant for the boda-boda sector. Studies indicate that minimizing idle time is a key requirement for drivers, as income is directly linked to vehicle availability (Simwaba & Qutieshat, 2026). BSS addresses this need by allowing a full "refueling" process within a few minutes, making it comparable to conventional refueling of internal combustion engine vehicles (Dash & Bandivadekar, 2021).

From a technical perspective, BSS require standardized battery designs, and communication protocols to ensure compatibility between vehicles and swapping stations. In addition, centralized charging infrastructure is required, where batteries are charged, monitored, and managed before being redistributed to users (Chen *et al.*, 2022). This enables better control over battery health and charging conditions but also introduces additional system complexity.

However, the deployment of swapping infrastructure requires significant capital investment, including station installation, big battery inventory, and operational logistics (Chen *et al.*, 2022). Additionally BSS presents more challenges. These include the need for standardization between manufacturers and the logistical complexity of managing large batteries fleets. Further the system relies on a reliable power supply, and if the battery fleet is too small, it can lead to long wait times at battery-swapping stations or a lack of fully charged batteries (Tarar *et al.*, 2023).This spotty availability of adequate and accessible charging and BSS infrastructure causes range anxiety concerns, which reduce the uptake of EV (Mwangi *et al.*, 2025).

Overall, BSS represents a viable solution for accelerating the adoption of e-motorbikes in Nairobi, particularly in high-utilization use cases. However, its scalability and cost-effectiveness depend on the ability to standardize systems and efficiently manage infrastructure and battery assets.

1.1.2 Pay as you go

The pay-as-you-go (PAYG) model is a widely used business approach that enables access to products or services through incremental payments rather than requiring a high upfront investment. The function of the PAYG system is explained in Figure 1.1. Instead of purchasing an asset outright, users pay in small amounts based on usage or over time, which lowers financial barriers and increases accessibility, particularly in low- and middle-income contexts (Barry & Creti, 2020).

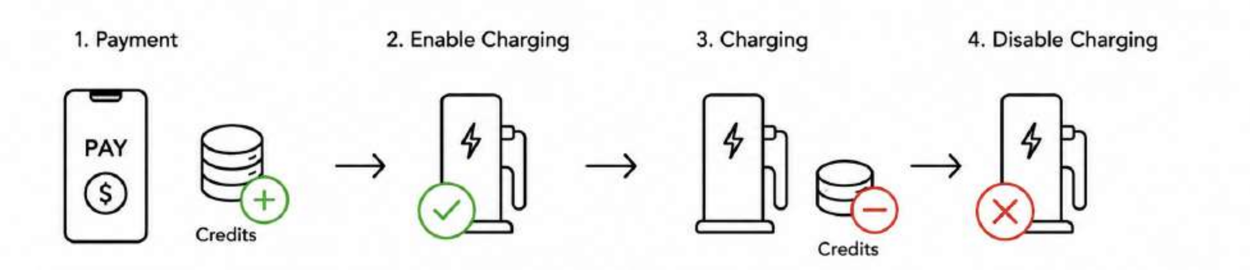

Figure 1.1. PAYG explanation

PAYG systems are commonly implemented in sectors such as energy access, where they are used to provide decentralized services, for example in solar home systems. In such applications, users make payments via mobile platforms, and access to the service is controlled accordingly, often through remote monitoring and control systems (Adwek *et al.*, 2019).

In Nairobi, where mobile payment systems such as M-Pesa are widely adopted, PAYG models can be seamlessly integrated into existing financial ecosystems. Compared to BSS, PAYG systems require less physical infrastructure, as they rely on controlled charging rather than battery exchange.

1.2 Payment in Kenya

Kenya is a country with extensive experience in digital financial services, primarily due to the rapid adoption of mobile money systems. Traditional banking infrastructure has historically had limited reach, especially among low- and middle-income populations. As a result, mobile-based payment solutions have become the dominant method for financial transactions, particularly in urban areas such as Nairobi as well as in rural regions (Mbiti & Weil, 2015).

The most prominent mobile payment platform in Kenya is M-Pesa, which was introduced in 2007 by Safaricom. M-Pesa allows users to deposit, transfer, and withdraw money using mobile phones, without requiring access to a traditional bank account. Over the years, it has evolved into a comprehensive financial ecosystem, supporting services such as bill payments, merchant transactions, and micro-financing. In Nairobi, mobile payments are deeply integrated into everyday life. Transactions such as public transport fares, food purchases, and utility payments are frequently conducted via mobile platforms (Mbiti & Weil, 2015). Recent reports indicate that a large majority of the population in Kenya actively uses mobile money, with penetration rates exceeding 80% and more than 95% per cent of loans issued digital (Sommer *et al.*, 2025).

1.3 Value Chain

The value chain of electric mobility in Kenya is characterized by a strong reliance on international supply, particularly for key components such as batteries and electronic systems. Battery technologies are typically imported, with a large share originating from manufacturers in China (Merino, 2026).

Vehicles such as e-motorbikes and e-bikes are often not imported as fully assembled units. Instead, they are delivered as kits and companies like eWAKA or ROAM assemble them locally. This approach reduces import costs and allows companies to adapt products to local requirements while creating local employment opportunities (Dankers, 2024).

In Nairobi, local assembly and distribution play a central role in the value chain, linking international suppliers with end users. However, the dependence on imported components remains a key factor influencing costs, availability, and scalability of electric mobility solutions. (Dankers, 2024)

1.4 Open APIs

Application Programming Interfaces (APIs) are a fundamental concept in modern software systems, enabling communication and data exchange between different applications or system components. An API defines a set of rules and protocols that specify how software entities can interact with each other. By providing standardized interfaces, APIs allow systems to exchange information without requiring direct access to their internal implementation (Ofoeda *et al.*, 2019).

APIs are commonly used to connect frontend applications, backend services, and external platforms. In this context, they act as intermediaries that process requests and return structured responses, usually in formats JSON or XML (Guido Barbaglia, 2017). One widely used architectural style is the Representational State Transfer (REST) approach, where communication is based on standard HTTP methods such as GET, POST, PUT, and DELETE. This approach enables scalable and platform-independent system integration (Ofoeda *et al.*, 2019).

A key characteristic of APIs is their ability to abstract complexity. Instead of requiring developers to understand the full internal logic of a system, APIs expose only the necessary functionalities. This facilitates the integration of third-party services, such as payment platforms or data providers (Mohammed Mudassir & Mohammed Mushtaq, 2024). In addition, APIs support modular system design, allowing individual components to be developed, tested, and maintained independently.

Open APIs, in particular, are publicly accessible interfaces that enable external developers to interact with a system or service. They are often accompanied by documentation and authentication mechanisms to ensure secure access. By enabling interoperability between different systems, open APIs play a crucial role in the development of scalable and flexible digital infrastructures (Neumann *et al.*, 2021).

Overall, APIs provide the technical foundation for connecting distributed systems and enabling interactions between software components. Their standardized structure make them a key enabler for integrating diverse functionalities within a unified system architecture.

1.5 Justification and Research Questions

1.5.1 Justification

The adoption of electric mobility in Nairobi is constrained by a combination of economic, infrastructural, and technical challenges. High upfront costs, limited charging infrastructure, and uncertainties related to battery usage and financing models hinder the large-scale transition from ICE vehicles to electric alternatives. While concepts such as BSS and PAYG systems have been proposed to address these barriers, there remains a need for technically feasible and scalable solutions that can be implemented within the existing local context.

In particular, the widespread use of mobile payment systems such as M-Pesa provides a unique opportunity to integrate financial transactions directly into the operation of charging infrastructure. However, the practical implementation of such systems requires reliable communication between payment platforms, backend systems, and physical charging hardware. Ensuring that charging processes can be dynamically controlled based on confirmed payments represents a key technical challenge.

The goal of this work is therefore the development of a PAYG battery charging system in collaboration with eWAKA. The system is intended to function as a scalable prototype that integrates a mobile payment mechanism and enables controlled charging based on payment status. The development process includes both laboratory testing under controlled conditions and subsequent field-oriented evaluation to assess its practical applicability.

1.5.2 Research Question

Based on this motivation, the following research question is formulated:

> *What scalable and cost-effective solution allows the control of battery charging processes based on payment information from an M-Pesa account using open APIs?*

To answer the research question, several technical approaches for enabling charging processes had to be investigated and developed. Therefore, additional research objectives were defined to structure the implementation and evaluation of the proposed system. In collaboration with the company eWAKA, this work focuses on the design, implementation, and evaluation of a PAYG charging architecture for electric two-wheelers in Nairobi. Particular attention is given to the integration of mobile payment systems, backend infrastructure, and charging control mechanisms based on open APIs and existing smart battery systems

The research objectives of this work are summarized as follows:

- Design a PAYG battery charging system suitable for the eWAKA fleet.
- Develop and implement a functional prototype of the proposed system.
- Integrate payment, logging, and monitoring functions into the prototype.
- Connect the developed solution to the existing eWAKA environment and operational infrastructure.
- Perform field tests to evaluate system availability, reliability, and practical feasibility under real operating conditions.

2 METHODS

2.1 Architecture of the PAYG system

This chapter describes the methodological approach used for the development of the PAYG charging system. The Figure 2.1 illustrates the architecture used to implement the PAYG system. The system was developed using an iterative approach, with continuous testing, evaluation, and refinement throughout the project.

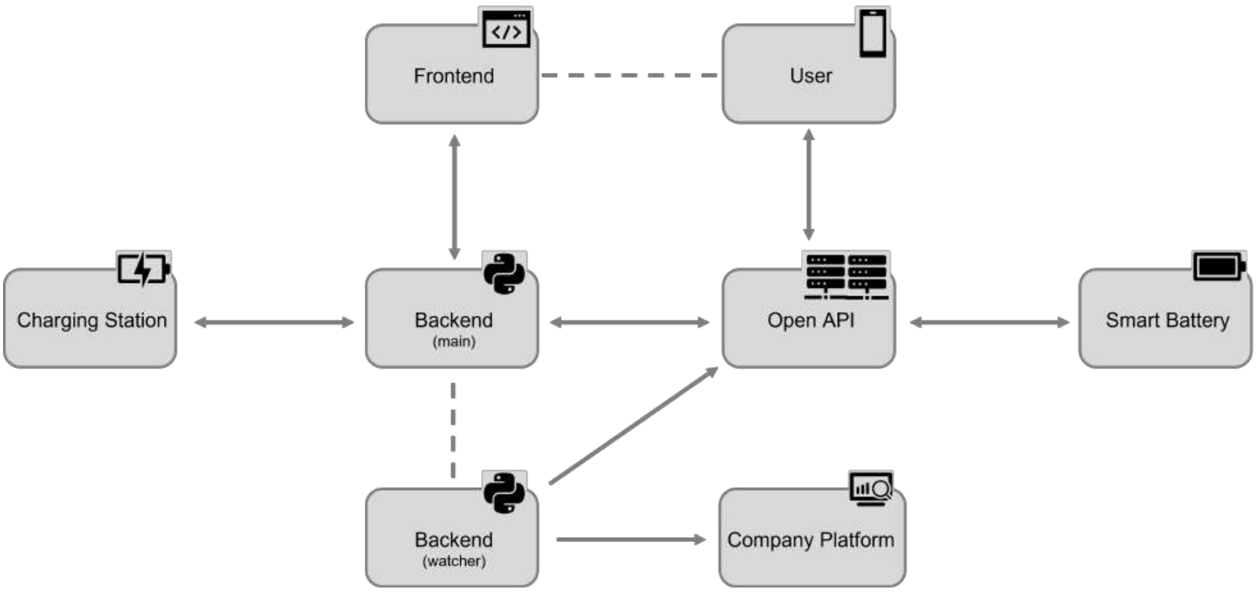

Figure 2.1. Architecture overview of PAYG system

The charging station is a hardware-based charging system for e-bikes developed as a prototype for this project. It includes a mobile network communication module that enables data exchange with the server. The server hosts two backend scripts and provides the connection to the frontend interface. In addition, the server communicates with two open APIs. The first API is the Daraja API, which is used to send payment requests directly to the user's mobile phone via M-Pesa. The second API is provided by the battery manufacturer and enables remote battery monitoring and control. The smart battery used in this project was mainly used for e-motorbikes and was already available within the company. It includes integrated internet communication capabilities and can be accessed through the manufacturer's open API. Furthermore, the company platform is already used internally to monitor the vehicle fleet and visualize operational data.

The next chapters are structured around the individual components of the PAYG architecture. They describe the development approach for the frontend, backend, payment integration, and charging enabling logic. The charging enabling logic section also covers the development of the charging prototype for e-bikes and the communication with the smart e-motorbike batteries.

2.2 Frontend

The development process was initiated with the user interface (UI). This part of the system could be implemented independently of a virtual private server (VPS), allowed an early start without generating cost for hosting, and made it possible to establish an initial user interaction concept before the backend infrastructure and server environment were fully defined. The frontend therefore served as an accessible starting point.

At the beginning, the UI was based on HTML and designed around the minimum information required to initiate a charging session. The payment process was expected to be based on the M-Pesa mobile payment system; therefore, the user interface included an input for the user's phone number, as well as a four-digit identifier or twelve-digit battery number used to assign the request to a specific charging system. These parameters formed the basis for the early interaction logic and allowed the charging session to be linked to both a payer and a physical charging point.

As the project evolved, the frontend was further refined. In this context, an additional selection parameter was introduced in the form of charging tiers. Three distinct charging tiers were defined, allowing users to select a predefined amount of energy to charge. The choice of three options in the prepaid model ensured technical feasibility and economical flexibility. A fully dynamic charging process with postpaid billing was not pursued, as it would have complicated the payment process and communication between the backend and the charging station. A limited number of options also enabled the use of the decoy effect to highlight a dominant choice and still provide the user with a certain degree of flexibility (Bateman *et al.*, 2008).

During the iterative development process, both the specific values and the interpretation of the charging tiers were adjusted multiple times in response to technical constraints and the evolving understanding of user requirements. In a later phase of the project, the concept of user interaction was adjusted to fit the operating environment of the company. Therefore, the original HTML-based standalone frontend was replaced with a WhatsApp-based UI and integrated into the already used WhatsApp interface of the company.

2.3 Backend

The backend was developed following the initial frontend design in order to enable data processing, system control, and communication between the different components. The backend served as the central element for linking user input, payment information, and charging information. Python was chosen as the programming language, while the Flask Web framework was used to implement a lightweight and modular web application.

In the early stages of development, the backend was hosted locally and exposed to the internet using the Ngrok reverse proxy tool, which provided an internet tunneling solution. This approach enabled system testing without the need for a fully deployed server infrastructure. In particular, it allowed for the reception of external API callbacks, which were required for the integration of the payment functionality.

In a later stage of the development on the VPS, an additional software script and a SQLite database was introduced. The database was designed to store and manage relevant information for each charger or battery, such as the selected charging tier, the amount of energy charged, and the current state of charge (SOC). The software module was implemented as a Python-based program running on the backend and was designed to continuously monitor the charging status of all connected chargers or batteries.

As the project progressed and the system requirements became more defined, the need for a more stable and permanent deployment environment emerged. At this stage, the backend was migrated to a VPS, which provided continuous availability and allowed the system to operate independently. This transition marked a shift from a purely experimental setup towards a more application-oriented system configuration.

2.3.1 Payment function through open API

It was decided to use the M-Pesa mobile payment system for payment processing. To enable communication with the M-Pesa platform, the frequently used Daraja API was selected. This API provides access to core payment functionality and supports automated transaction handling. Within the Flask route in the backend, payment requests were sent to the M-Pesa platform through this API. This involved sending a structured request containing relevant parameters such as the user's phone number, the tier and the transaction amount. As a result, the M-Pesa platform generates a payment prompt on the user's mobile device, which requires confirmation of the transaction. The API subsequently provides information on the outcome of the payment, allowing the system to determine whether the transaction was successful or not.

This response information was sent by callbacks to a specified URL. Therefore, the backend was made accessible via a public endpoint, and a Flask route was created to handle the incoming response.

2.4 Enabling Charging

To enable charging for both e-bikes and e-motorbikes, two different methodological approaches were followed. The development initially focused on the e-bike use case. In this context, a charging station was selected as the central element for controlling the charging process. The underlying concept included the creation of a connection between the charging station and the backend through a SIM module, allowing the system to verify the payment status and, accordingly, enable or disable charging. The actual control of the charging current was implemented at the battery level. This required the integration of an IoT component within the battery system, which could receive control signals from the charger. To enable this communication, an additional information cable was installed in parallel to the power cable.

At an early stage of development, the activation and confirmation of the charging process were implemented using SMS communication. This approach was chosen due to its simplicity and broad availability and was realized using the Africa's Talking API. Charging commands and status updates were exchanged between the charging station and the backend via text messages. However, during the iterative development process, this method revealed limitations in terms of cost, latency, and system complexity. As a result, the communication approach was revised, and the system was adapted to use mobile data instead of SMS. Charging activation and status updates were then handled via HTTP-based requests sent directly from the charging station to the backend. Although a secure HTTPS connection would have been preferable, implementation challenges led to the temporary use of an open HTTP port for transmitting confirmation messages.

In a later stage of the project, the focus was extended to e-motorbikes. In contrast to the e-bike setup, these systems were found to use batteries with integrated communication capabilities. This made it possible to interact directly with the battery via an existing API provided by the manufacturer, removing the need for a custom-built charging station. From a methodological perspective, this represented a shift from a hardware-based to a software-based integration approach, significantly simplifying the system design and enabling rapid scalability across multiple batteries.

3 Results and discussion

This chapter presents the results of the iterative development process and describes the final version of the PAYG components. Figure 3.1 provides a detailed overview of the information flow between the different components of the PAYG system.

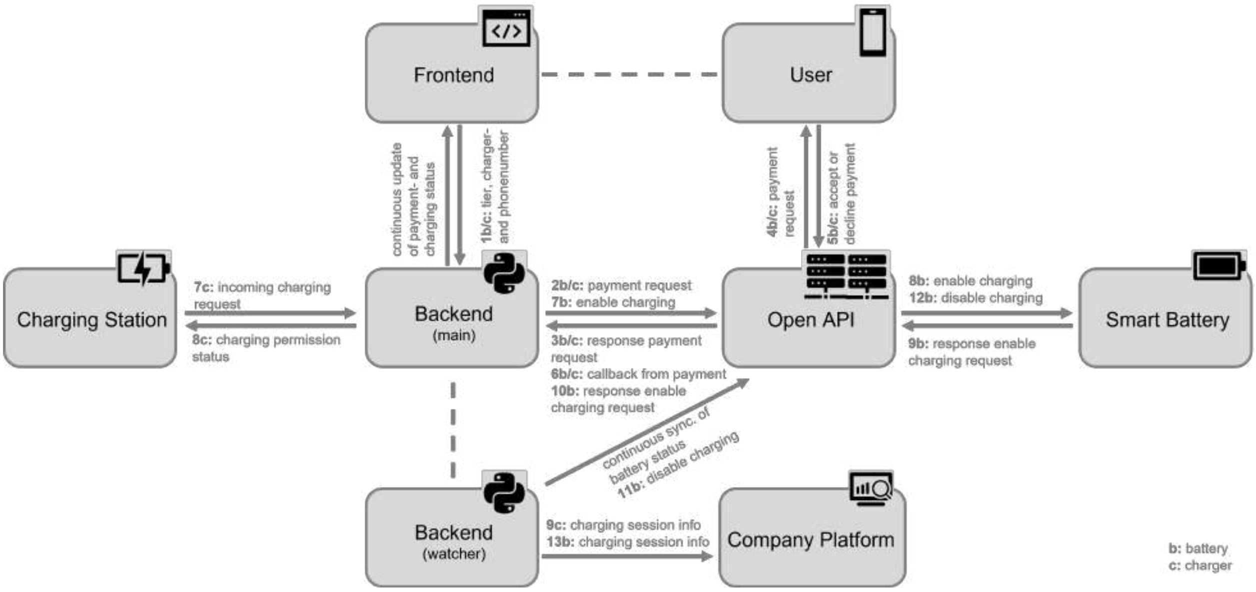

Figure 3.1. Informationflow overview of PAYG system

The following chapters describe the final version of the frontend, backend, and charging enabling logic. The frontend chapter presents the final UI and the selected parameters. The backend section explains the software structure and the detailed implementation of the code. The complete source code is available online via the project's GitHub repository [GHE GitHub]. The charging enabling section presents the final prototype of the developed e-bike charging station and demonstrates the interaction with smart batteries for charging e-motorbikes.

3.1 Frontend

The frontend was designed to transmit either a 12-digit battery number or a 4-digit charger number, together with a 12-digit phone number, to the backend. The charger number was used to initiate a charging session for e-bikes with the corresponding prototype charging station, whereas the battery number was required to start a charging session for an e-motorbike battery. The phone number was required to send the payment push request to the user's mobile device. In addition, the desired amount of energy to be charged had to be transmitted so that the corresponding price could be requested. Three charging levels were identified as the most suitable for the business model, corresponding to charging levels of 20%, 70%, and 100%, with associated prices of 100, 190 and 240 Kenyan Shillings. The user behavior indicated a preference for smaller charges. However, since frequent low SOC levels can negatively impact battery health, the pricing structure was adjusted to make smaller charging tiers less economically attractive. Different UI were evaluated for this purpose. Initially, an HTML-based web interface was developed, followed by the implementation of an interactive WhatsApp-based interface.

3.1.1 HTML solution

The HTML UI shown in Figure 3.2 was implemented to allow users to enter the required information. By selecting a charging tier button, the UI successfully transmitted the input data to the backend. The charging tier could only be selected after all user inputs had been successfully validated. As long as the input data did not have the correct number of digits or did not begin with a valid sequence of digits, the field was highlighted in red. Once the validation process was completed, the three green buttons became active and were used to transmit the information to the backend. If the required inputs were incomplete or invalid, the buttons remained disabled and appeared faded. In addition, an information field provided real-time feedback to guide the user through the process and indicate the required actions.

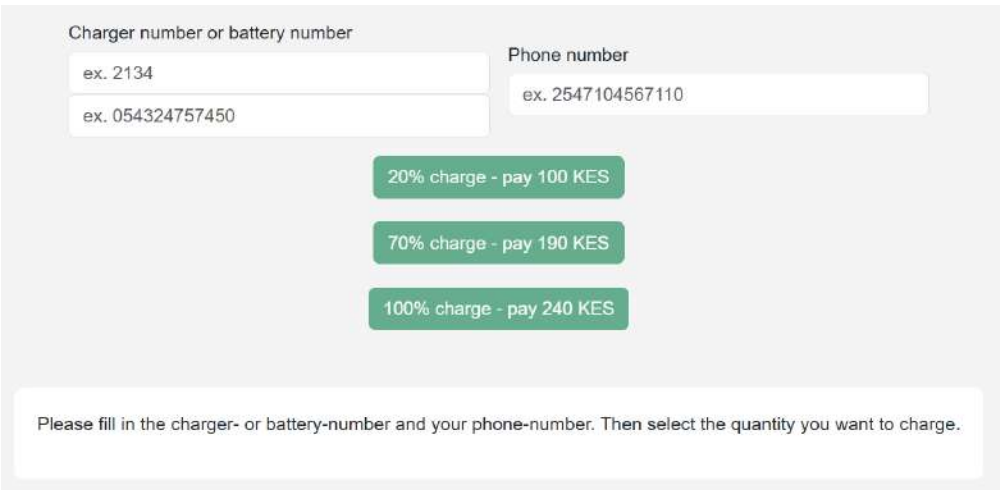

Figure 3.2. HTML UI

3.1.2 WhatsApp solution

In a later stage, the frontend functionality was successfully integrated into the existing WhatsApp-based platform of the company, as shown in Figure 3.3. The WhatsApp user interface is activated by sending a message to a designated company phone number. This initiates an interactive dialog, which allows the user to select from various available functions. These functions included confirming arrival at a pickup location, collecting an order, and confirming a completed delivery. Since drivers were already using this interface for operational tasks, the addition of a charging function enabled a seamless transition without the need for a separate application. Integration was achieved through an external development partner, which had already developed the previous WhatsApp features. As a result, the existing HTML UI logic had to be reproduced and transferred to the WhatsApp interface.

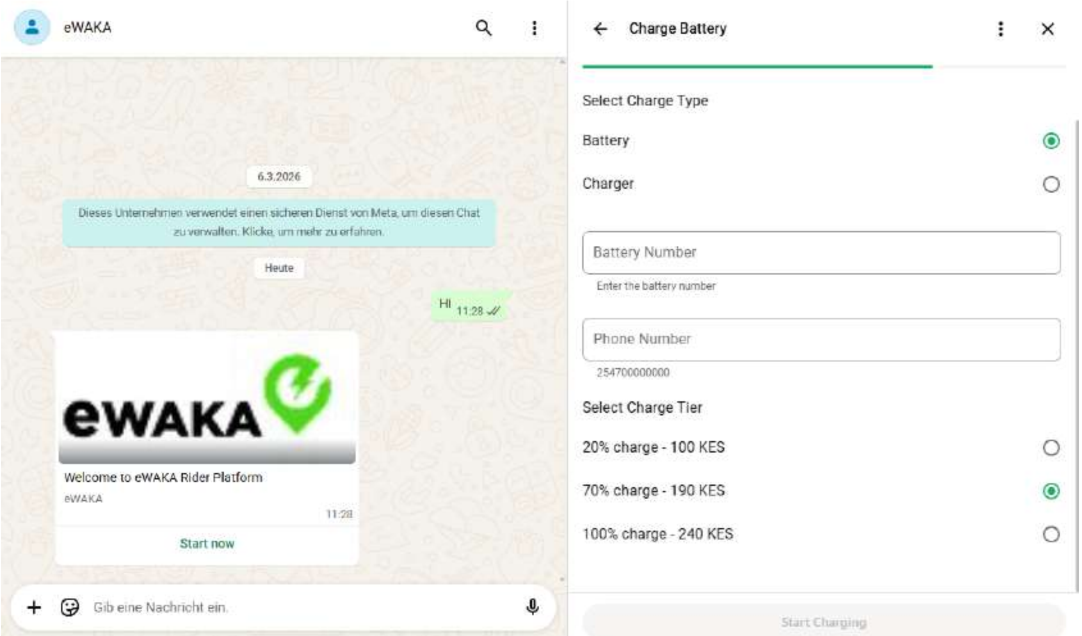

Figure 3.3. WhatsApp UI

3.1.3 Comparison of HTML and WhatsApp UI

During use, clear differences between the two UI became apparent. The HTML-based interface provided specific advantages and disadvantages compared to the WhatsApp interface. Table 3.1 presents the advantages and disadvantages of the HTML interface, while Table 3.2 summarizes the characteristics of the WhatsApp-based interface.

Advantages	Disadvantages
Usable on both smartphones and computers	Purchase and maintenance of a domain
Clear user guidance through information field	

Table 3.1. Advantages and disadvantages of the HTML UI

Advantages	Disadvantages
Users are already familiar with the platform	Dependence on a third-party provider for development
Potentially lower mobile data usage due to free WhatsApp access packages	Reduced visualization compared to a graphical web interface

Table 3.2. Advantages and disadvantages of the WhatsApp UI

3.2 Backend

The backend of the developed system was implemented as a Python-based application, with the core logic centralized in a main script. This script integrates a set of modular helper functions. Among the key functionalities is a payment push function, which initiates payment requests via the Daraja API. In addition, authentication functions were implemented to generate and manage API tokens required for secure communication with external platforms. Further backend functions were developed to interact with API enabled batteries. These include the activation and deactivation of the charging ability, the retrieval of information about the state of the battery, such as the SOC, and the detection of alarm or error states. The backend also supports the transmission of relevant operational data of each charging session to an external company platform for further analysis and monitoring.

To structure system entities, object-oriented models were introduced in the form of classes representing chargers and batteries. These classes allow the assignment and management of specific attributes, enabling scalable handling of multiple devices. Sensitive information such as credentials, API keys, and URLs are managed through environment variables to ensure secure configuration.

As the system evolved from a prototype to a scalable solution, a database was introduced to manage charging sessions. A SQLite database is automatically generated by the main script and is used to store information such as the payment status and the charging status for each charger and battery. Initially, no database was required due to the limited scope of a single prototype.

However, with increasing system complexity and the need to manage multiple simultaneous charging sessions, persistent data storage became essential. The database structure includes separate tables for chargers and e-motorbike batteries.

The backend logic is executed through multiple routes defined within the Flask web application. These routes manage communication between the frontend, backend, charging stations, and external APIs. Table 3.3 provides an overview of the implemented Flask routes and their primary functions.

Flask Route	**Description**
`/Save/Input`	Receives and validates user inputs from the frontend, including charger numbers, battery numbers, phone numbers, charging tiers, and rental parameters. The validated data is then stored in the database and forwarded to the backend logic.
`/Payment/Push/Request`	Initiates payment push requests through the Daraja API and stores the generated checkout request identifier required for matching asynchronous payment callbacks.
`/mpesa/callback`	Receives asynchronous payment callbacks from the M-Pesa API, evaluates transaction success or failure, updates the database accordingly, and enables or blocks charging processes.
`/Incoming/Charging/Request`	Handles communication requests from charging stations. The route verifies the payment status of a charger and returns the corresponding charging permission and charging tier information.
`/Payment/Status`	Provides real-time payment status updates for chargers and smart batteries to the frontend interface.
`/Charger/Status`	Provides the current charging status of chargers and smart batteries, allowing the frontend to monitor active charging sessions.

Table 3.3. Overview of implemented Flask routes

During operation, it was observed that charging sessions may need to be interrupted due to work environments, such as incoming tasks for drivers. To address this, the system was extended to track the amount of charging credit already used, allowing sessions to be paused and resumed without loss of paid value. Unfortunately, this allowed users to charge small again and unintentionally encouraged users to keep batteries at low SOC levels. Therefore, in addition to the main script, a separate module referred to as the SOC-watcher was developed. This Python-based process continuously monitors active charging sessions and battery states. Its primary function is to ensure that charging is stopped once the amount of paid energy has been reached. Furthermore, it monitors battery alarms and error conditions and attempts to handle them where possible. This component enables continuous monitoring of the system beyond event-based triggers.

Additional operational features were introduced to support the usage of the system. Specific phone numbers were implemented to allow internal company charges at a minimal cost, enabling operational charges without standard user pricing. In addition, a timestamp was added to each charge request, allowing the system to track the last usage of each battery. This made it possible to identify inactive batteries and support follow-up actions, such as user notification, to prevent long-term battery degradation. For deployment, the application was hosted on a virtual private server (VPS). The Flask application was served using Gunicorn, while Nginx was configured as a reverse proxy to manage incoming web traffic and forward requests to the application server. The VPS was configured with open ports for both HTTP and HTTPS communication, with HTTP access also supporting communication with the charging station modules. Version control and collaboration were managed by GitHub, where the project is publicly available and can be updated.

Finally, the backend system is connected to a company-specific platform which aggregates and visualizes system data. This platform presents information received from the backend and the SOC-watcher in a structured and accessible format, enabling company staff to monitor system performance, analyze charging behavior, and maintain an overview of all active devices and sessions.

3.3 Charging station for e-bikes

After several iterations, a functional prototype of a charging station for e-bikes was successfully implemented. The system architecture consists of a primary processing unit within the charging station and a secondary control unit integrated into the battery. It was observed that the business operates with a variety of batteries and chargers featuring different connectors and specifications (see Appendix A.4). To address this variability, a modular system was realized in which future adaptations can be done. The design was initially focused on a battery type for which sufficient technical information was available. Strong emphasis was placed on low-cost realization, achieved through a 3D-printed enclosure, the use of standard electronic components, and the integration of an existing manufacturer-provided charger.

3.3.1 Primary processing unit

The primary processing unit consisted of a micro controller, a SIM module for communication with the backend, a Hall-effect sensor for current measurement, and a dedicated power supply. The wiring diagram of the primary processing unit is shown in Figure 3.4 and images of the final prototype are visible in the Appendix A.1.

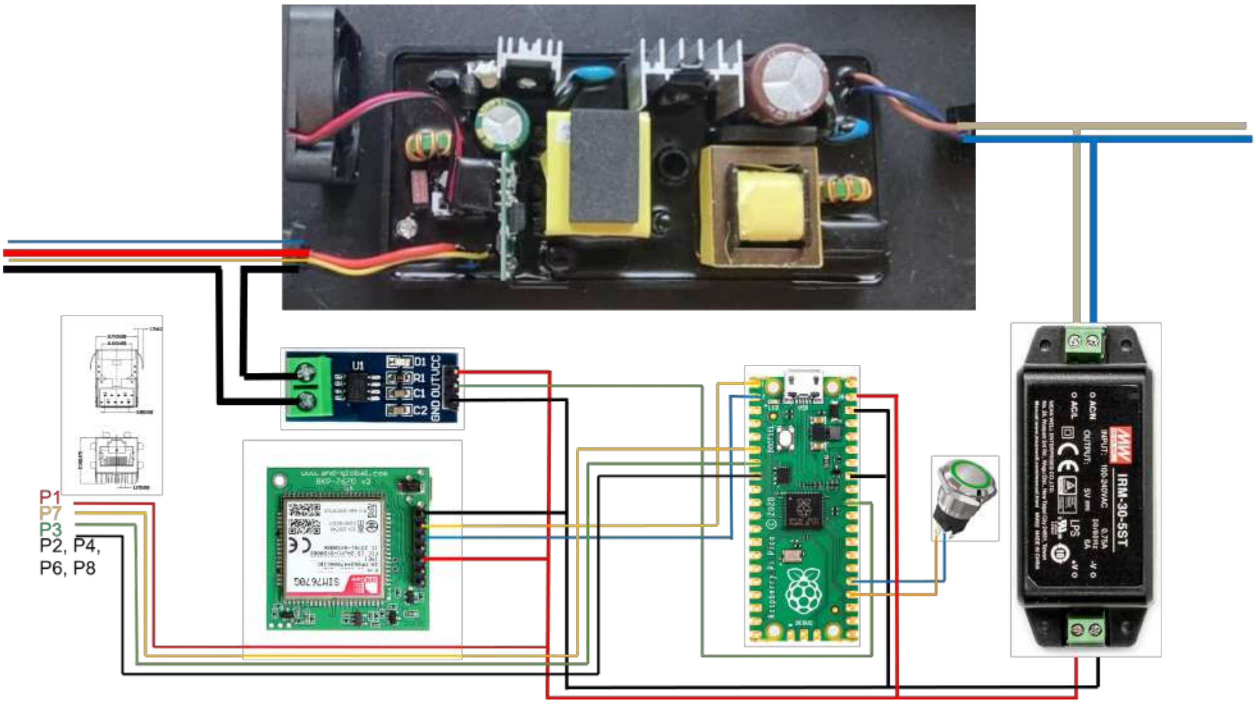

Figure 3.4. Wiring of primary process unit

The operational behavior of the primary processing unit was successfully demonstrated. Upon connecting the charging station to a power source, the system was automatically initialized and powered the SIM module, which was indicated by the flashing button three times. After initialization, the system remained in a waiting state until user interaction occurred by pressing the button. Once activated, the system began sending periodic HTTP POST requests to the backend to retrieve the current payment status. This functionality was shown to operate reliably, allowing the system to determine whether charging should be enabled. Following a positive response, communication with the secondary control unit was established, and the charging process was successfully activated. The transferred current was measured and converted to the corresponding energy under the assumption of constant 58 V output of the charger.

The SIM-based communication was successfully implemented and enabled the charging station to establish a mobile data connection and exchange information with the backend. To communicate with the SIM module, AT commands were used, allowing retrieval of network status, signal quality, registration information and more. Reliability of operation was observed to require correct registration of the SIM module within the mobile network. Not all IoT SIM cards were identified to support mobile data by default. Proper configuration with the network provider was therefore necessary to ensure successful packet-switched data communication. Due to connection and certificate-related issues, it was not possible to establish a HTTPS connection. Therefore, HTTP communication was used instead. Since the charging station only polls the charging authorization and no sensitive payment information through this connection, this does not represent a critical security risk. Although a potential man-in-the-middle attack within the mobile network cannot be entirely excluded, it is considered highly unlikely. In such a case, an attacker could at most grant unauthorized charging access, but no financial data would be compromised. Nevertheless, a switch to HTTPS should be considered for future scalability.

3.3.2 Secondary control unit

The secondary control unit inside the battery was implemented with simple logic. The wiring diagram is shown in Figure 3.5.

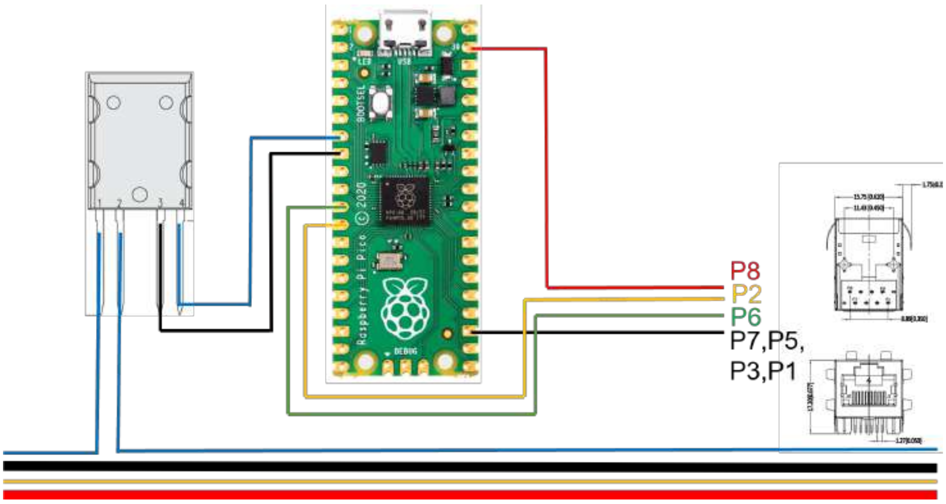

Figure 3.5. Wiring of secondary control unit

It is powered by the same connection used for serial communication and remains in a passive state until a command is received from the primary unit. Upon receiving an activation signal, the relay is switched to allow charging. Conversely, when no connection to the primary control unit exists, the relay blocks the charging process. This design also serves as a backup safety mechanism, ensuring that charging is only possible when the system explicitly authorizes it. It was found that integrating the relay directly into the main power line was not feasible due to the high voltages of 58 V and the requirement for galvanic isolation. Instead, the relay was successfully integrated into an existing low-voltage communication line, enabling safe and reliable switching. Serial communication between the primary and secondary units was shown to function reliably, enabling command transmission for both the activation and deactivation of charging.

3.3.3 Hardware integration of prototype

This chapter describes hardware integration of the developed charging station for e-bikes. Table 3.4 provides an overview of the main hardware components used in the primary process unit and the secondary control unit.

Primary Process Unit	**Secondary Control Unit**
Raspberry Pi Pico (RP2350)	Raspberry Pi Pico (RP2350)
SIM7670G communication module	CPC1786J solid-state relay
eWAKA charger (54 V, 3 A) see Appendix A.4	Ethernet female port
Push button (momentary self-reset)	Connection cables
Mean Well IRM-30-5ST 5V power supply (30W)	
Arduino Hall effect sensor	
Ethernet cable	
PLA (Bambulab PLA black)	
Connection cables	

Table 3.4. Hardware components used in the charging prototype

A 5V power supply was selected to ensure stable operation of all integrated components. Particular attention was paid to the SIM7670G communication module, as mobile communication modules can require short current peaks of up to 2 A during network communication. Therefore, a stable and sufficiently dimensioned power supply was necessary to ensure reliable operation. An Ethernet cable was selected as the serial communication cable between the primary and secondary control units. This solution simplified the wiring process and reduced concerns related to serial communication distances and signal reliability. Furthermore, the CPC1786J relay was chosen to provide galvanic isolation between the control electronics and the charging system. Since the relay was later integrated into an existing safety circuit, a simpler 5V relay solution would also have been technically sufficient for the application.

In addition, the prototype's housing was designed using CAD and then 3D-printed. Figure 3.6 shows the final version of the prototype housing. During the development process, the CAD design was iterated several times, as individual hardware components and system requirements changed throughout the project. The final CAD files of the charging station are publicly available on the project GitHub repository [GHE GitHub].

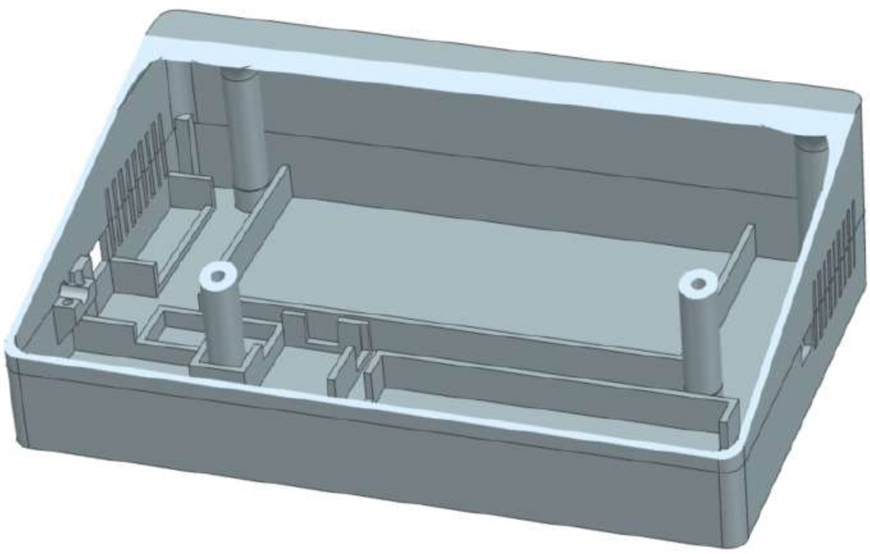

Figure 3.6. CAD of prototype housing

3.4 Charging for e-motorbikes

As identified during the development process, the batteries used in e-motorbikes could be directly controlled via an API provided by the battery manufacturer. This enabled the activation and deactivation of the charging and discharging ports without the need for additional IoT in the battery. The API could be accessed either through a dedicated web interface provided by the manufacturer or directly via the backend system. The web interface from the battery manufacturer is shown in the Figure 3.7. The interface provides battery information on voltage and error status. After successful integration of the API with the backend, it was possible to trigger the charging process immediately after a confirmed payment and to automatically deactivate the charging once the available credit was consumed. This demonstrated that a fully software-based control of the charging process is feasible, making the approach highly scalable and cost-efficient, as existing hardware infrastructure can be utilized without modification.

Figure 3.7. Battery web interface allowing manual control of the battery charging and discharging through the buttons marked in red

3.4.1 Example charging session

The functionality of this approach is illustrated in Figure 3.8, which shows the progression of a charging session over time. The SOC is colored blue, while the current flow is colored red, distinguishing between the charging and discharging phases. The displayed data corresponds to a battery that was used during the day, resulting in a low initial SOC, followed by multiple charging and discharging cycles over time.

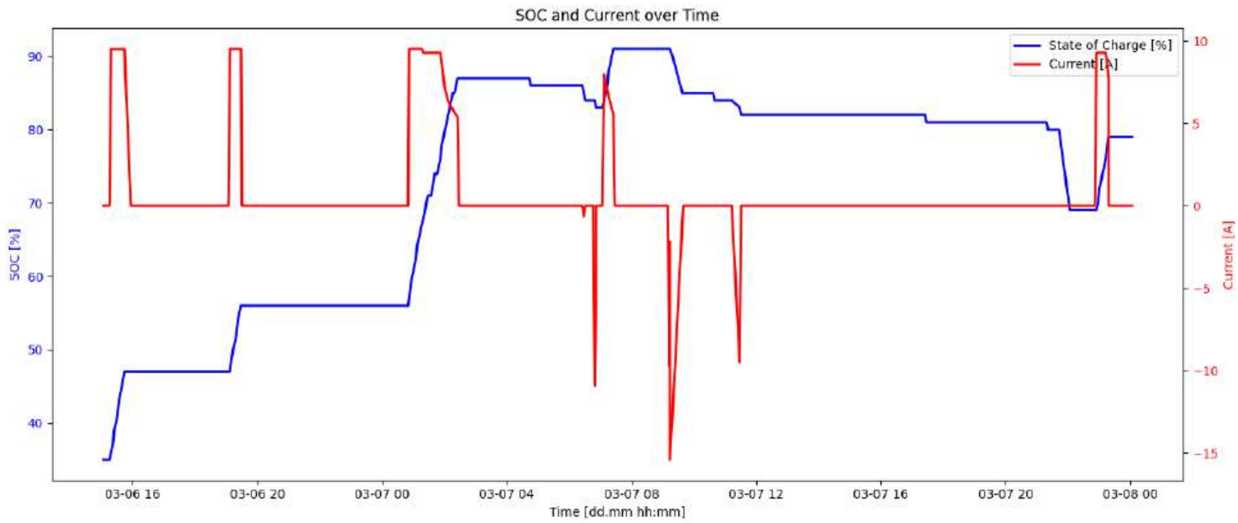

Figure 3.8. SOC history of a 70% charging session

The analysis of the graph provides several relevant observations. First, it was found that the cumulative increase in SOC during charging session corresponds to the total purchased charging credits. In the presented case, the sum of all positive SOC increments results in an overall increase of 70%, which matches the value of a full charging session. This confirms the correct implementation of credit-based charging control. It can also be observed that discharge cycles do not affect the charging session

Furthermore, the graph reveals characteristic battery behavior. At higher SOC levels, a slight decrease in SOC can be observed even without active discharging, indicating self-discharge effects. In addition, the data shows irregularities in the recording intervals. During the final discharge phase, no discharge current is recorded, suggesting incomplete data acquisition in this segment. Another observable trend is the relationship between charging current and SOC. As the SOC increases and the cell voltage rises, the charging current decreases. This behavior is consistent with typical battery charging characteristics and further supports the validity of the recorded data.

3.5 Discussion

This work demonstrates that a PAYG charging system for electric two-wheelers can be technically implemented by mobile payment. The results highlight two fundamentally different approaches for enabling charging, each with distinct implications for scalability, cost, and system complexity.

3.5.1 Comparison of Charging Approaches

The hardware-based solution for e-bikes confirms that a PAYG mechanism can be realized independently of existing battery intelligence. The developed prototype successfully demonstrates the feasibility of such a system in a controlled environment. However, this approach is currently limited to a prototype stage and is not yet suitable for large-scale deployment.

For scalability, the hardware design would require significant refinement. The current system, which relies on discrete electronic components, would need to be consolidated into a dedicated printed circuit board (PCB) to reduce cost, size, and assembly complexity. In addition, the enclosure would need to be redesigned for mass production. These steps require both financial investment and specialized engineering expertise, which may not be available internally at eWAKA and would likely need to be sourced externally.

In contrast, the API-based solution for e-motorbikes eliminates the need for additional hardware entirely, as it utilizes existing communication capabilities of smart batteries already deployed by the company in combination with the PAYG API software. This significantly reduces both upfront investment and operational complexity. From a cost-benefit perspective, this approach is clearly advantageous, as it allows rapid scaling without modifications to the physical charging infrastructure.

However, this approach introduces a strong dependency on the battery manufacturer. The functionality of the system relies on the availability and stability of the manufacturer's API, as well as the embedded IoT hardware within the battery. It must be ensured that both current and future batteries support this interface. Furthermore, issues such as communication failures or unstable connections can affect the availability of the PAYG system. While the backend can be designed to handle such uncertainties robustly, the reliability of the overall system remains partially dependent on third-party technology.

These findings suggest that while the hardware-based approach provides independence and flexibility, it is associated with higher complexity and cost. The API-based approach, on the other hand, offers superior scalability and efficiency but at the expense of increased external dependency. In the context of this project, the API-based solution appears to be the more viable pathway for large-scale implementation.

3.5.2 Limitations and Practical Considerations

The developed system was primarily validated as a functional prototype, with a focus on rapid development and proof of concept. As a result, several aspects relevant for real-world deployment remain only partially addressed.

In the hardware-based solution, electrical safety and standardization represent critical challenges. Charging systems must comply with established safety standards, including requirements related to insulation, connector types, and protection mechanisms. While these aspects were simplified during prototyping to enable rapid iteration, they become more important when transitioning to a product intended for widespread use. This introduces a trade-off between development speed and compliance, which must be carefully managed in future iterations.

Data reliability also emerged as a limitation. In the case of the API-based solution, this issue is closely linked to the dependency on the battery manufacturer, as data quality and availability are determined by the underlying IoT system.

More generally, the system has not yet been evaluated under extended real-world operating conditions. Factors such as long-term reliability, system availability, behavior under unstable network conditions, and user interaction patterns remain insufficiently explored.

3.5.3 Implications for Future Development

The results indicate that further development should focus less on expanding system functionality and more on validation under real operating conditions. A comprehensive field testing phase is essential to assess system performance, identify failure modes, and understand user behavior.

In particular, the system should be deployed with end users to collect detailed operational data, including battery usage patterns, charging behavior, and system reliability. This would enable the identification of limitations related to undercharging, overcharging, or intermittent connectivity. Continuous monitoring of deployed batteries is therefore critical, and dedicated resources may be required to track system performance and respond to emerging issues.

From a technical perspective, future development should prioritize the API-based approach. Given its scalability and lower cost, it represents the most promising pathway for implementation. This requires close collaboration with the battery manufacturer to ensure reliable communication, consistent API support, and high-quality data access. Strengthening this interface is more impactful than further investment in standalone charging hardware.

Overall, the transition from a functional prototype to a scalable solution depends primarily on system validation, operational robustness, and the reliability of external dependencies. Addressing these aspects will be essential for enabling a sustainable and widely deployable PAYG charging system.

4 Conclusions

This work demonstrates the technical feasibility of a PAYG charging system for electric two-wheelers in Nairobi by linking mobile payment information with charging control. A functional prototype was successfully developed, integrating a frontend interface, a backend system, and a mobile payment solution based on M-Pesa. The system enables charging processes to be activated and limited according to confirmed payments, thus providing a practical implementation of a usage-based charging model.

4.1 Research Question Evaluation

The findings of this work provide the following answer to the research question defined at the beginning of this study.

> The results of this work indicate that the most scalable and cost-effective solution for controlling charging processes based on M-Pesa payment information is an API-based architecture using existing smart battery communication systems. Compared to a fully hardware-based charging approach, the API-based solution reduces hardware complexity, lowers implementation costs, and enables easier integration into existing fleet management environments. However, the scalability and long-term reliability of this approach strongly depend on the stability and availability of external battery APIs and communication networks.

The conclusion of the defined research objectives can be summarized as follows.

- **Design a Pay-as-you-go battery-charging system fitting to eWAKA's fleet**

  A modular PAYG charging architecture was successfully designed for the eWAKA fleet. The developed system combines frontend interfaces, backend processing, payment integration, and charging control mechanisms. The architecture supports both e-bike charging stations and API-enabled smart batteries.

- **Create a prototype of the system**

  A functional prototype was successfully implemented. Two charging enabling approaches were investigated. The first approach consisted of a hardware-based e-bike charging station prototype, while the second approach used direct API communication with smart batteries. Both concepts demonstrated the practical feasibility of PAYG-controlled charging.

- **Implement a payment and logging function to the prototype**

  The system successfully integrated payment requests through the M-Pesa Daraja API. In addition, logging and monitoring functions were implemented within the backend and database system. Operational parameters such as payment status, charging status, battery information, and charging history could be stored and monitored.

- **Integrate the payment and logging function into the eWAKA environment**

  The backend and monitoring system were integrated into the existing eWAKA infrastructure. The API-based battery approach enabled interaction with existing smart battery systems and company monitoring platforms without requiring major hardware modifications.

- **Perform field tests to determine availability and reliability**

  Initial field tests confirmed the general functionality and availability of the developed system under real operating conditions. Communication between the frontend, backend, payment API, and charging control systems was successfully demonstrated. However, the tests also revealed limitations related to the reliability of the mobile network, API stability, and battery communication, indicating the need for further long-term testing.

4.2 Comparison of Charging Approaches

Two different approaches to enabling charging were investigated. The hardware-based solution for e-bikes confirmed that PAYG functionality can be implemented independently of existing battery systems. However, this approach remains at a prototype stage and would require substantial further development, particularly in terms of hardware integration, manufacturing, safety compliance, and long-term reliability before large-scale deployment would be possible.

In contrast, the API-based solution for e-motorbikes proved to be significantly more efficient and scalable. Using existing communication capabilities of smart batteries, the need for additional hardware could largely be avoided, resulting in a more streamlined and cost-effective implementation. At the same time, this approach introduces dependencies on the battery manufacturer, particularly regarding API availability, communication reliability, and long-term software support.

4.3 Limitations and Future Work

Overall, the results indicate that PAYG charging systems have a strong potential to support the adoption of electric mobility in Nairobi, especially when integrated with widely used mobile payment platforms. The API-based approach appears to be the most promising path for large-scale deployment, provided that reliable communication with battery systems can be ensured.

Future work should focus on extensive long-term field testing under real operating conditions. This includes evaluating system reliability, user acceptance, charging behavior, and battery usage patterns, as well as identifying limitations related to mobile connectivity and data availability. In addition, future developments should address cybersecurity, database scalability, and hardware robustness for commercial deployment scenarios. Close collaboration with battery manufacturers will remain essential to improve API stability, increase system robustness, and ensure long-term viability.

4.4 Final Remarks

In conclusion, this work provides a practical foundation for the development of scalable PAYG charging solutions for electric mobility applications in developing urban environments. The project demonstrates that the combination of mobile payment systems, cloud-based backend infrastructure, and smart battery communication can enable flexible and usage-based charging concepts. In addition, the work highlights the technical and organizational challenges that must be addressed to achieve reliable large-scale deployment in real-world environments. Furthermore, the project demonstrates how existing digital payment ecosystems can support the transition toward more accessible and flexible electric mobility solutions in developing urban regions.

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Appendix A

These images show the prototype of the PAYG system for e-bikes.

(a) PAYG system prototype during charging

(b) PAYG system protoype disassembled

(c) PAYG system with corresponding e-bike

(d) PAYG system protoype disassembled

Figure A.1. Prototype of PAYG system

The company eWAKA primarily used two different types of e-motorbike batteries. The e-motorbike battery 1.0 is not equipped with an API-enabled IoT.

(a) Battery 1.0

(b) Battery 2.0

The company eWAKA used various batteries with different chargers. These images provide an overview of the five different chargers. Charger C2 and C5 resp C2.2 and C5.5 were used to charge e-motorbike batteries. The prototype for the e-bike PAYG charging system was developed based on the charger C1 resp C1.1.

(a) C1

(b) C1.1

(c) C2

(d) C2.2

(e) C3

(f) C3.3

(a) C4

(b) C4.4

(c) C5

Figure A.4. Overview of all chargers used at eWAKA

ETH

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Die unterzeichnete Eigenständigkeitserklärung ist Bestandteil jeder während des Studiums verfassten schriftlichen Arbeit. Eine der folgenden zwei Optionen ist **in Absprache mit der verantwortlichen Betreuungsperson** verbindlich auszuwählen:

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Titel der Arbeit:

Development of a pay-as-you-go battery-charging system for electric mobility in Nairobi Kenya

Verfasst von:
Bei Gruppenarbeiten sind die Namen aller Verfasserinnen und Verfasser erforderlich.

Name(n):	**Vorname(n):**
Fässler	Patrick

Ich bestätige mit meiner Unterschrift:
- Ich habe mich an die Regeln des «Zitierleitfadens» gehalten.
- Ich habe alle Methoden, Daten und Arbeitsabläufe wahrheitsgetreu und vollständig dokumentiert.
- Ich habe alle Personen erwähnt, welche die Arbeit wesentlich unterstützt haben.

Ich nehme zur Kenntnis, dass die Arbeit mit elektronischen Hilfsmitteln auf Eigenständigkeit überprüft werden kann.

Ort, Datum	**Unterschrift(en)**
31.05.2026 Zürich	P. Fässler

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